

IoT and Embedded Systems

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Room – CS231

Network Layer

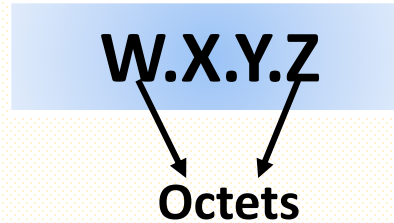
IPv4, Addressing, Routing

- ❑ Transport datagram (data of the transport layer) from a source to a destination node.
- ❑ Responsible for:
 - Routing
 - Breaks the packet into smaller packet is required.
 - Logical addressing.
- ❑ Network Layer provides unreliable service:
 - A packet may lost in transit
 - Packet may arrive out of order
 - Duplicate packets may be generated.

ABL ✓
✓ 00

- ❑ A unique address used by host/node for transmission of packet in internet.
- ❑ Why not simply use MAC address?
 - Because MAC address does not follow any hierarchy and thus a single packet can create congestion.
- ❑ Network Layer provides unreliable service:
 - A packet may lost in transit
 - Packet may arrive out of order
 - Duplicate packets may be generated.

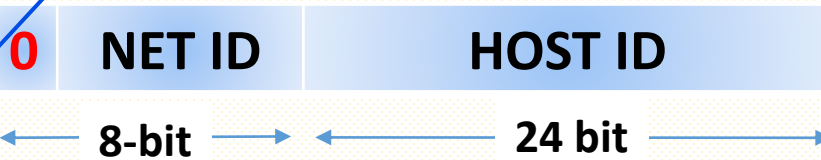
- ❑ IPv4 is a 32 bit quantity, expressed as a dotted decimal notation



- ❑ IPv4 Address: 01000010 10001000 00110000 01111110
(66) (134) (48) (126)
- ❑ Dotted decimal notation: 66.134.48.126

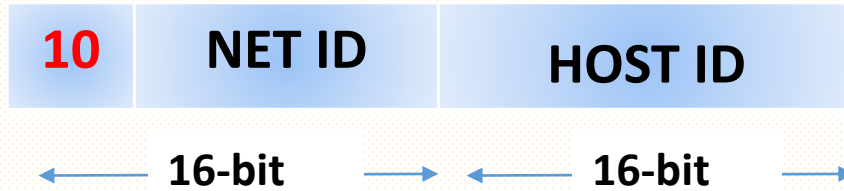
- ❑ It consist of two parts:
 - Network Number → Assigned and managed by central authority (IANA→RIR→ISPs)
 - A Host Number → Assigned and managed by local network administrator. (LAN)
 - During routing only network part is used.

Class A



1.0.0.0 to 127.255.255.255

Class B



128.0.0.0 to 191.255.255.255

Class C



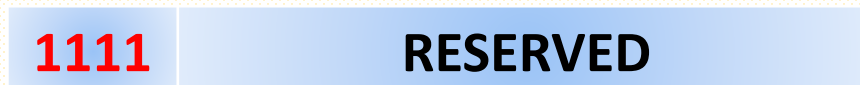
192.0.0.0 to 223.255.255.255

Class D



224.0.0.0 to 239.255.255.255

Class E



240.0.0.0 to 255.255.255.255

❑ Which class does this IP address belongs to?

- 10.0.0.1
- 172.16.254.1
- 150.100.10.5
- 201.45.67.89
- 65.10.5.2
- 192.0.0.1

❑ Convert the first octet to its binary form and check the starting bit(s):

- i.e. 10.0.0.1 $\rightarrow (10)_{10} = (00001010)_2$: (Starts with 0 $\rightarrow$ Class A)
- i.e. 150.100.10.5 $\rightarrow (150)_{10} = (10010110)_2$: (Starts with 10 $\rightarrow$ Class B)

Maximum Network/Host in a Class



❑ Class A

- Maximum Network Possible: $2^7 - 1 = 127$ (126 usable)
- Maximum Host: $2^{24} - 2 = 16777214$

❑ Class B

- Maximum Network Possible: $2^{14} - 1 = 16383$
- Maximum Host: $2^{16} - 2 = 65534$

❑ Class C

- Maximum Network Possible: $2^{21} - 1 = 2097151$
- Maximum Host: $2^8 - 2 = 254$

❑ Why (-2) and (-1):

- First Address: Network Number
- Last Address: Broadcast Address

❑ Network Address

- NETID. HOSTID (All 0)
- i.e. Class A (X.0.0.0) → 40.0.0.0
- Class B (X.Y.0.0) → 129.55.0.0
- Class C (X.Y.Z.0) → 192.168.5.0

❑ Direct Broadcast Address

- NETID.HOSTID(All 1's)
- i.e. Class A (X.255.255.255) → 40.255.255.255
- Class B (X.Y.255.255) → 129.55.255.255
- Class C (X.Y.Z.255) → 192.168.5.255

Special Purpose IP



☐ For Private Use:

- 10.x.x.x
- 172.16.x.x to 172.31.x.x
- 192.168.x.x

☐ Local/Loopback

- 127.0.0.0 to 127.255.255.255

☐ Default N/W

- 0.0.0.0

☐ Limited Broadcast

- 255.255.255.255

IPv4 Packet Format

Header
20 to 60 Byte

Payload (Layer 4 Data)
0 to 65535 – Header Size

0100

Version (4-bit)	HLEN (4-bits)	Type of Service (8-bit)	Total Length (16-bits)	
Identification (16-bit)			Flag 3-bit	Fragment Offset 13-bit
TTL (8-bit)	Protocol (8-bit)		Header Checksum 16-bits	
Source Address (32-bit)				
Destination Address (32-bit)				
Options (40 Byte Maximum; Optional)				
Data (Payload)				

IPv4 Packet Headers

- ❑ Version (0-3): Version of the IP packet format
 - IPv4 → 0100
 - IPv6 → 0110
 - IPv5 → (Experimental version; not used)
- ❑ HLEN (4-7): Internal header length (in 32 bit word)
 - Minimum: 20 byte (HLEN Value → 0101) ~~×~~ 
 - Maximum: 60 byte (HLEN Value → 1111)
- ❑ Type of Service (8-15): It is replaced by DSCP (Differentiated Service Code Point) and ECN (Explicit Congestion Notification) (6+2)bits
 - 101110 (VoIP, real time apps)
 - 100010 (Streaming Media)
 - 011010 (Video Conference)

IPv4 Packet Headers

- ❑ Total Length (16-31): Total length of the packet (Header + Data)
 - Max IPv4 packet size? Length = 16-bits $\rightarrow 2^{16} \rightarrow 65536$ bytes
- ❑ Identification (32-47): unique ID for each fragmented packet
- ❑ Flag (48-50): Flags to control fragmentation.
 - DF $\rightarrow$ Don't fragment
 - MF $\rightarrow$ more fragment
- ❑ Fragment Offset (51-63): Position of fragment (in 8-byte blocks)
 - Why Fragmentation needed?
 - IP Packet $\rightarrow 65536$ Bytes
 - MTU of Ethernet $\rightarrow 1500$ Bytes
 - MTU of WiFi $\rightarrow 1500$ Bytes
 - MTU of LoWPAN $\rightarrow 128$ Bytes

IPv4 Packet Headers

- ❑ Time To Live (64-71): Max hope (how many routers this packet can cross). At each hope the value will be decremented by 1. (avoid loops).
- ❑ Protocol (72-79): Upper layer protocols
 - ICMT(1)→IGMP(2)→UDP(6)→TCP(17) (Protocol Name (number) with lower to higher priority)
- ❑ Header Checksum (80-95): For error detection in header part.
- ❑ Source and Destination Address (96 to 159) : 32-32 bit source and destination address
- ❑ Options (Max 40 bytes)
 - Route record: IP address of routers b/w SA and DA
 - Source routing: if source want to follow the exact route (then can specify here)

Questions



☐ An IPv4 packet has a header length of 40 bytes. What is the value of the HLEN field in the IPv4 header?

40 / 5

☐ An IPv4 packet has an HLEN value of 5 and carries 1000 bytes of data. What is the total length of the IPv4 packet?

1020

☐ If the HLEN field has a binary value of 0110, what is the header length in bytes?

24

☒ What is the maximum amount of data (payload) that can be carried in an IPv4 packet?

65535

☐ If an IPv4 packet's Total Length field is 1500 and the HLEN field is 5, What is the size of the payload?

1480

Fragmentation

❑ Why needed?

- Each layer imposes some maximum size of packets, due to various reasons. Called Maximum Transfer Unit (MTU).

❑ Suppose a large packet travels through a network whose MTU is too small.

- Fragmentation (and also reassembly) is required.
- Each fragment is transmitted as a separate IP packet.
- Fragmentation is typically done by routers.
- Fragments reassembled later: **transparent** or **non-transparent**.

FRIDR

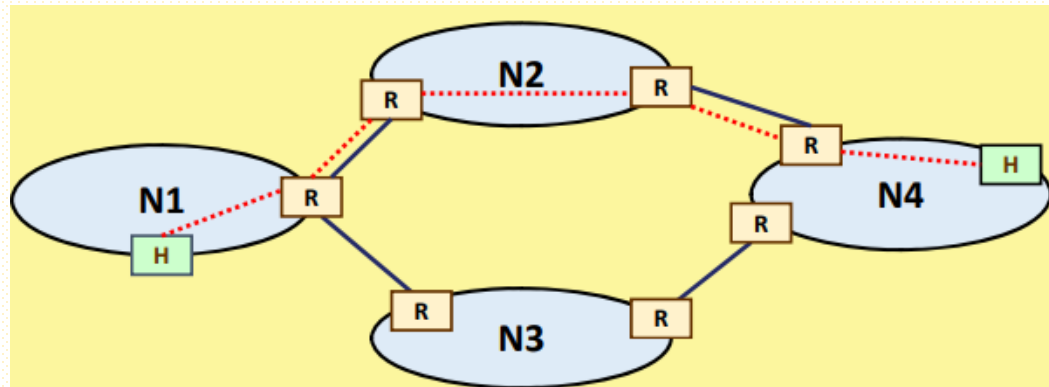
Fragmentation

❑ Why needed?

- Different network types (Ethernet, Wi-Fi, etc.) have different MTU values. If a router encounters a packet larger than the MTU of the outgoing link, it must fragment it so that each fragment fits.

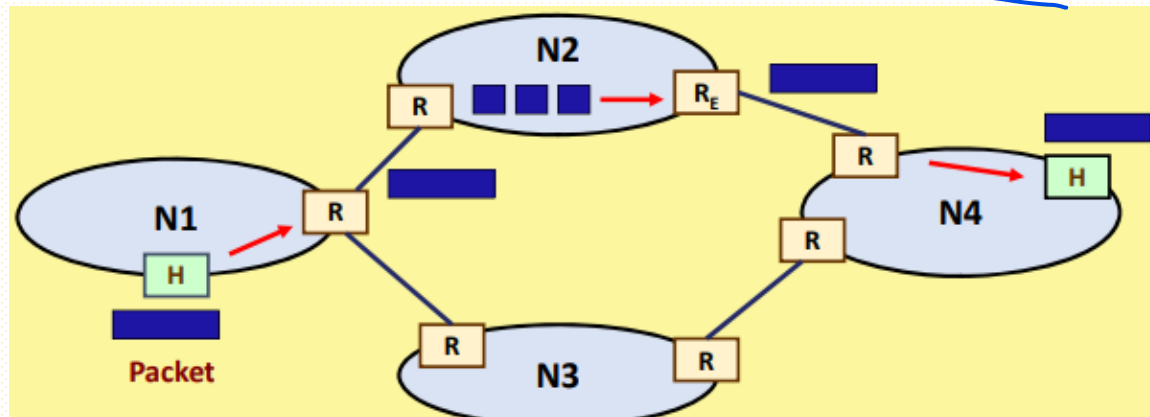
❑ Suppose a large packet travels through a network whose MTU is too small.

- Fragmentation done, each fragment is transmitted as a separate IP packet. Fragmentation is typically done by routers.
- Fragments reassembled later: **transparent** or **non-transparent**.



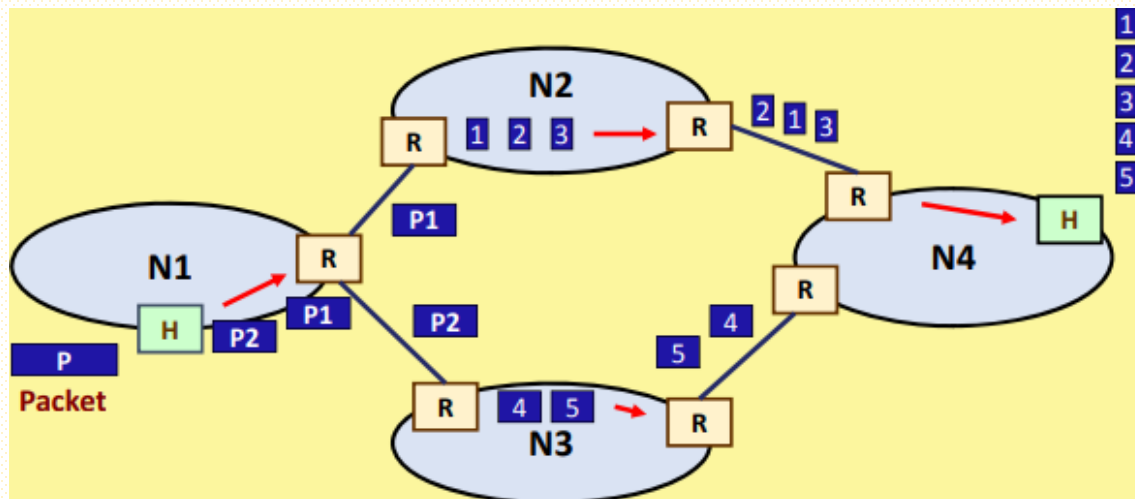
Transparent Fragmentation

- ❑ Fragmentation and reassembly is done entirely within a single network. (fragmentation is hidden from destination or other network).
- ❑ Entry router of the network fragments the packet and exit router reassembles it before forwarding to other router.
- ❑ Not used in modern network (Was used during ARPANET):
 - Not suitable for large or dynamic network (higher memory uses, follow same exit router)
 - Packet may get fragmented and reassembled repeatedly.



Non-Transparent Fragmentation

- ❑ Fragmentation can take place at any router and will not be resembled by any intermediate router. (used in IPv4)
- ❑ Each fragment is treated as an independent packet and are resembled at destination host.
- ❑ Challenge:
 - A single fragment loss? → packet loss.
 - Reassembly delay at destination.
 - Each fragment (minimum 20 byte overhead)



Fragmentation

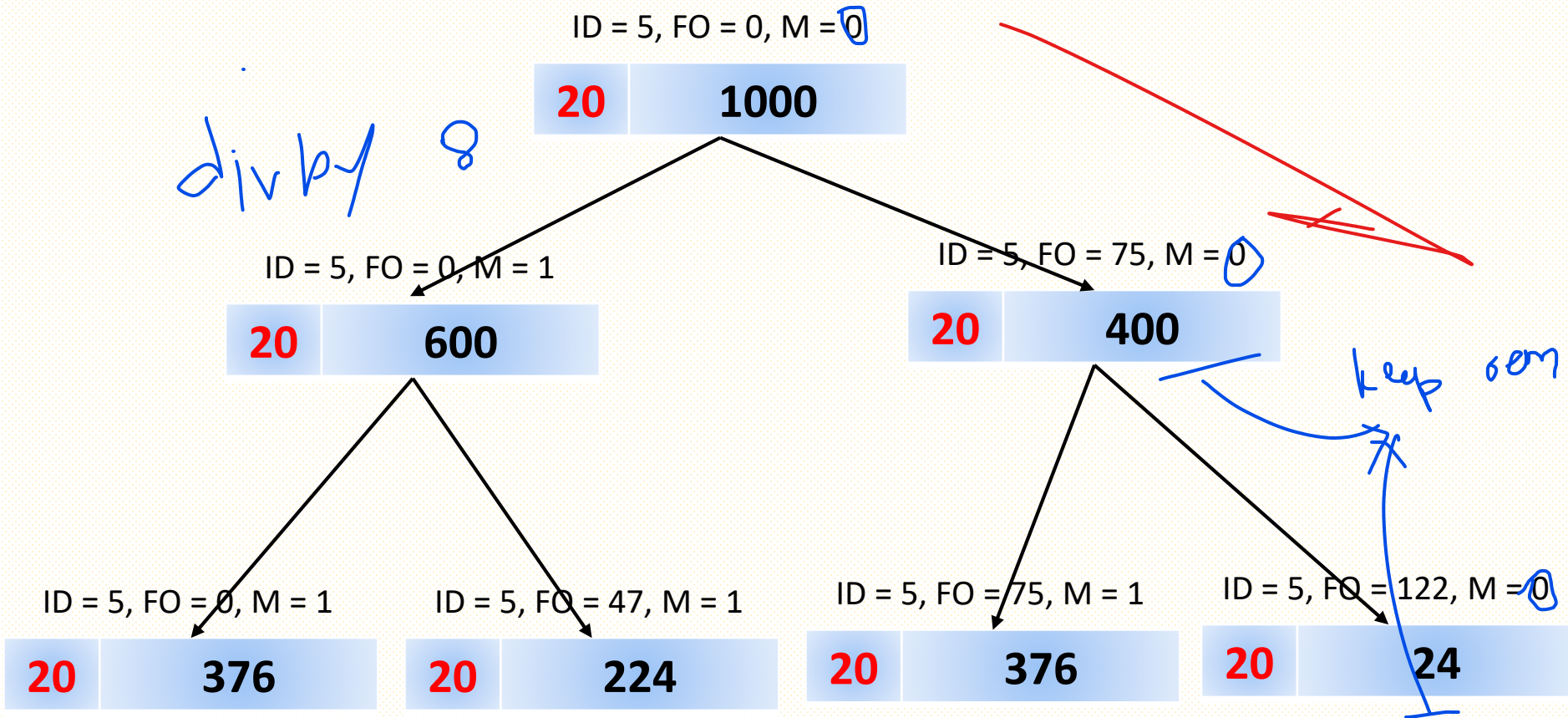


Identification (16-bit)	Flag 3-bit	Fragment Offset 13-bit
----------------------------	---------------	---------------------------

- ❑ Identification (16-bits): an ID set by the source.
- ❑ Fragment offset (13-bit): Indicates where in the original datagram this packet belongs to (specified in multiple of 8 bytes)
- ❑ Flags (3-bits):
 - D-bit (don't fragment): prevent fragmentation from taking place.
 - M-bit (more fragment): specifies if this fragment is the last one or not.
- ❑ What if D=1 and MTU of the network is small?
 - Router will drop the packet.
 - ICMP packet to host → Type 3, Code 4, with MTU size.

Fragmentation: Example

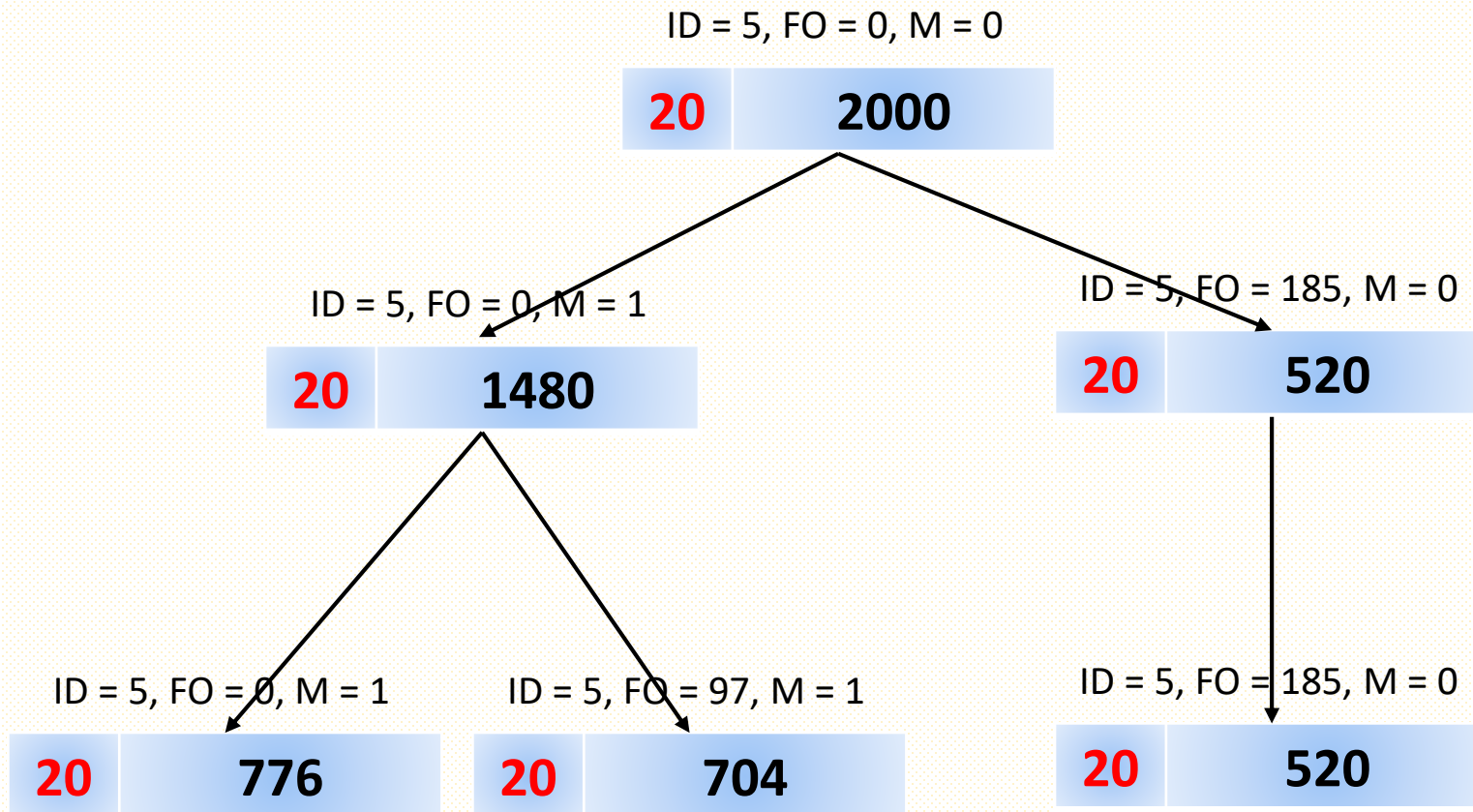
❑ 1000 byte data $\rightarrow$ N1 (MTU: 620Bytes) $\rightarrow$ N2(MTU:400 Bytes).



❑ 1020 Byte sent $\rightarrow$ 1080 Byte received

Fragmentation: Example

❑ 2000 byte data $\rightarrow$ N1 (MTU: 1500Bytes) $\rightarrow$ N2(MTU:800 Bytes).



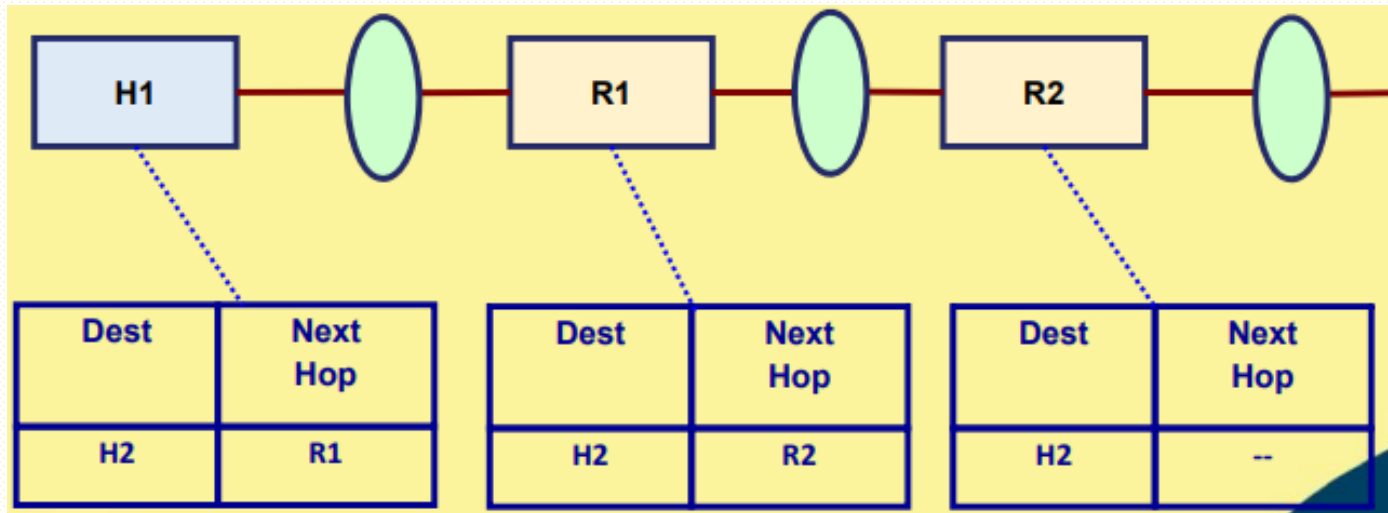
❑ 2020 Byte sent $\rightarrow$ 2060 Byte received

Routing (IPv4)

- ❑ A process of selecting a path for traffic in an network.
- ❑ Packet delivery options:
 - Direct Delivery: Directly from host-to-host, or router-to-host.
 - Indirect Delivery: Through one or more routers.
- ❑ Routing decision is taken based on:
 - Routing Table (Static/Dynamic)
 - Routing Algorithms (RIP, BGP etc.)

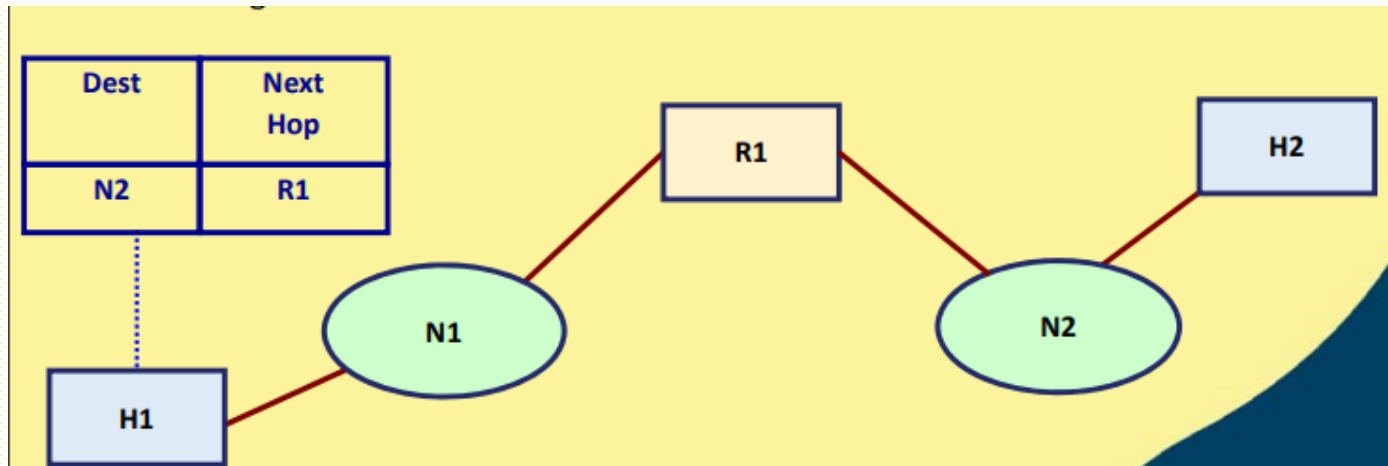
Routing Types

- ❑ Next-Hop Routing: Routing table stores the route of next hop (router address to which the packet should be sent).



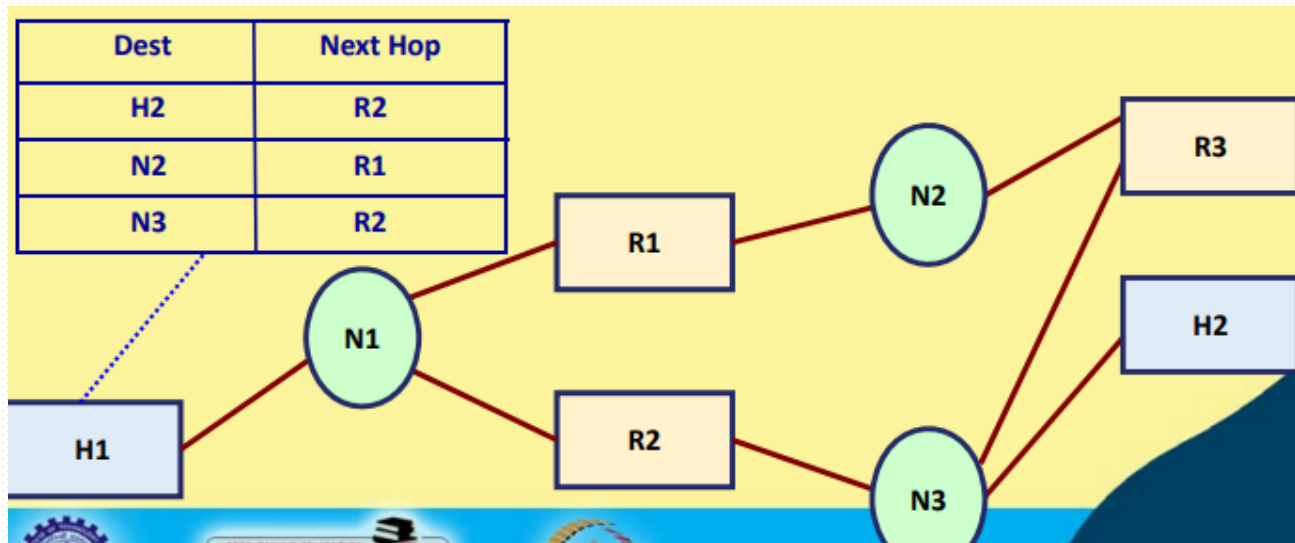
Routing Types

- ❑ Network-Specific Routing: Routers routing table contains route of entire network.



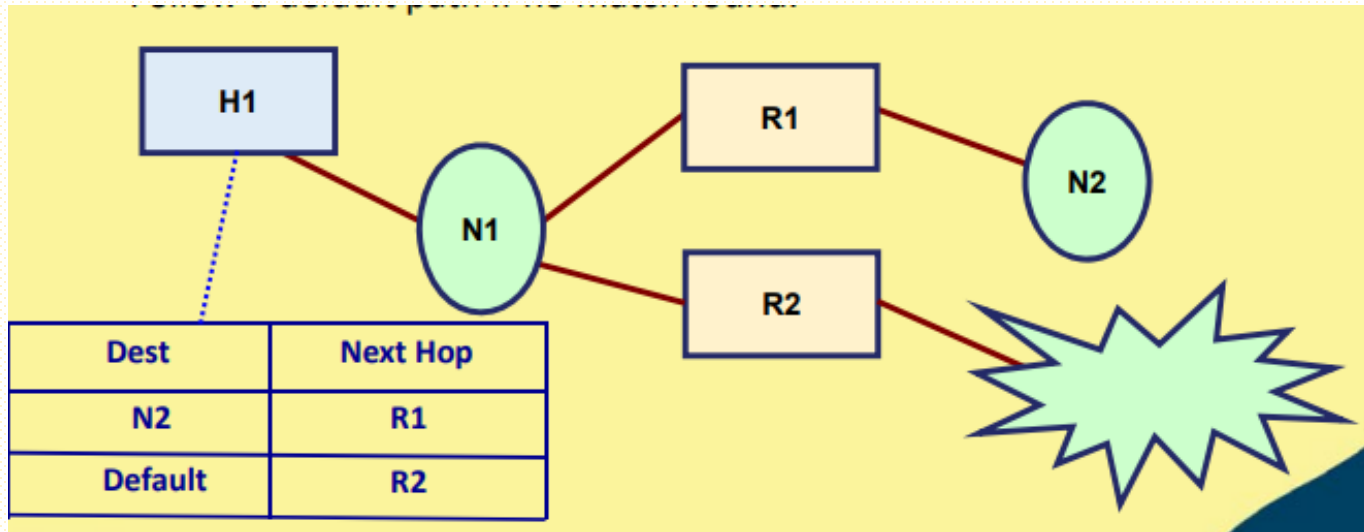
Routing Types

- ❑ Host-Specific Routing: Routes are defined for individual IP address



Routing Types

- ❑ Default Routing: Follow a default path if no route found.



NNHD

Routing Table

- Subnet Mask (determines the portion of network in IP address)
- Destination IP Address (destination n/w or host)
- Next Hop Address (address of next router)
- Flags:
 - U: Router is up and running.
 - G: Destination is in another network (go via next hope).
 - H: Host-Specific Address (route is for specific host, not a network).
 - D: Added by redirection (route was added by ICMP redirect).
 - M: Modified by redirection (route was modified by ICMP redirect).
- Interface (specifies local interface i.e. eth0, wlan0)

DGNFI

Destination	Gateway	NetMask	Flag	Interface
192.168.1.0	0.0.0.0	255.255.255.0	U	Eth0
10.10.0.0	192.168.1.1	255.255.0.0	UG	Eth0
172.16.5.20	192.168.1.254	255.255.255.255	UGH	Eth0
0.0.0.0	192.168.1.1	0.0.0.0	UG	Eth0

Routing Table

- ❑ On Unix/Linux System → `netstat -r`
- ❑ On windows System → `route print`

```
(kali㉿kali)-[~]
$ netstat -r
Kernel IP routing table
Destination      Gateway          Genmask          Flags      MSS Window  irtt Iface
default          10.57.34.223    0.0.0.0          UG         0 0        0 eth0
10.57.34.0       0.0.0.0         255.255.255.0    U         0 0        0 eth0

(kali㉿kali)-[~]
```

```
IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          10.57.34.223     10.57.34.43      35
10.57.34.0                  255.255.255.0    On-link          10.57.34.43      291
10.57.34.43                 255.255.255.255  On-link          10.57.34.43      291
10.57.34.255                255.255.255.255  On-link          10.57.34.43      291
127.0.0.0                   255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                   255.255.255.255  On-link          127.0.0.1        331
127.255.255.255             255.255.255.255  On-link          127.0.0.1        331
169.254.0.0                 255.255.0.0      On-link          169.254.245.78   281
169.254.245.78              255.255.255.255  On-link          169.254.245.78   281
169.254.255.255             255.255.255.255  On-link          169.254.245.78   281
```

Routing Table

```
IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          10.57.34.223     10.57.34.43      35
10.57.34.0                 255.255.255.0    On-link          10.57.34.43      291
10.57.34.43                255.255.255.255  On-link          10.57.34.43      291
10.57.34.255               255.255.255.255  On-link          10.57.34.43      291
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        331
127.255.255.255           255.255.255.255  On-link          127.0.0.1        331
169.254.0.0                255.255.0.0      On-link          169.254.245.78   281
169.254.245.78            255.255.255.255  On-link          169.254.245.78   281
169.254.255.255           255.255.255.255  On-link          169.254.245.78   281
```

- ❑ Network Destination: The destination IP address or subnet to match.
- ❑ Netmask: Used to define the subnet size.
- ❑ Gateway: Where the traffic is forwarded if the destination matches.
- ❑ "On-link": means it's directly reachable.
- ❑ Interface: The local IP used to send the packet.
- ❑ Metric: Cost or priority of the route; lower means preferred.

Routing Table

```
IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          10.57.34.223     10.57.34.43      35
10.57.34.0                  255.255.255.0    On-link          10.57.34.43      291
10.57.34.43                 255.255.255.255  On-link          10.57.34.43      291
10.57.34.255                255.255.255.255  On-link          10.57.34.43      291
127.0.0.0                   255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                   255.255.255.255  On-link          127.0.0.1        331
127.255.255.255             255.255.255.255  On-link          127.0.0.1        331
169.254.0.0                 255.255.0.0      On-link          169.254.245.78   281
169.254.245.78              255.255.255.255  On-link          169.254.245.78   281
169.254.255.255             255.255.255.255  On-link          169.254.245.78   281
```

- ☐ 0.0.0.0 0.0.0.0 10.57.34.223 (Default route): The packet will be forwarded to gateway 10.57.34.223.
- ☒ Local Network (10.57.34.0/24), On-link (directly reachable) → don't go via gateway. Broadcast address → 10.57.34.255.
- ☐ Loopback Address: 127.x.x.x

Routing Table

```
(kali㉿kali)-[~]  
$ netstat -r  
Kernel IP routing table  
Destination      Gateway          Genmask          Flags      MSS Window  irtt  Iface  
default          10.57.34.223    0.0.0.0          UG          0 0        0     eth0  
10.57.34.0       0.0.0.0         255.255.255.0    U           0 0        0     eth0  
  
(kali㉿kali)-[~]
```

- ❑ Destination: The destination subnet or default route.
- ❑ Gateway: Next-hop IP address; where packets should go
- ❑ Genmask: Subnet mask (for matching destination)
- ❑ Flags: Routing flags
- ❑ Iface: Interface to use (e.g., eth0)

Sub-Nets

Subnet Mask

- ❑ A subnet mask is used to determine network and host portion.
- ❑ i.e. 11111111.11111111.11111111.00000000
 - All 1's → network part
 - All 0's → represent host part
- ❑ Default subnet mask (for classful addressing):

Class	IP Range	Mask	n/w	Host
A	0.0.0.0 – 127.255.255.255	255.0.0.0	128	16M
B	128.0.0.0 – 191.255.255.255	255.255.0.0	16384	65000
C	192.0.0.0 – 223.255.255.255	255.255.255.0	2M	254

Subnet Mask



Class	IP Range	Mask	n/w	Host
A	0.0.0.0 – 127.255.255.255	255.0.0.0	128	16M
B	128.0.0.0 – 191.255.255.255	255.255.0.0	16384	65000
C	192.0.0.0 – 223.255.255.255	255.255.255.0	2M	254

- ❑ 128 network means → 128 unique network ID.
- ❑ 16 M host → you can connect 16M unique systems/devices within a single network ID.
 - 1.0.0.0 → can connect 16M systems
 - 2.0.0.0 → can connect 16M systems.
 - ~ 2.1B devices can be connected with Class A IP's.

Subnet Mask



Class	IP Range	Mask	n/w	Host
A	0.0.0.0 – 127.255.255.255	255.0.0.0	128	16M
B	128.0.0.0 – 191.255.255.255	255.255.0.0	16384	65000
C	192.0.0.0 – 223.255.255.255	255.255.255.0	2M	254

❑ What could be the problem if we use such huge network?

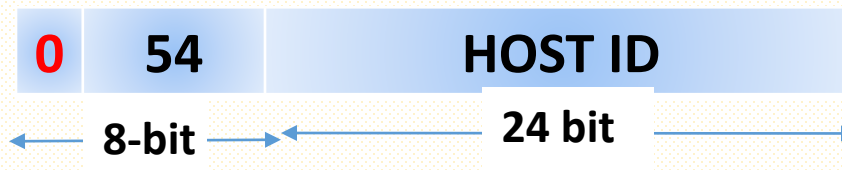
- Broadcast
- Latency
- Congestion
- Underutilization

B L W

❑ Sub-netting is used to solve that issue.

Sub-netting

- ❑ Allow a block of address to be split into several parts for internal use as multiple networks.
- ❑ The network will still act as single network for outside the world.
- ❑ Sub-net introduces a third level hierarchy N/W|SUB-NET|HOST
- ❑ It borrows a portion of the host space and applies it to network space.
- ❑ Consider a network 52.0.0.0; it can have $2^{24} - 2$ host.

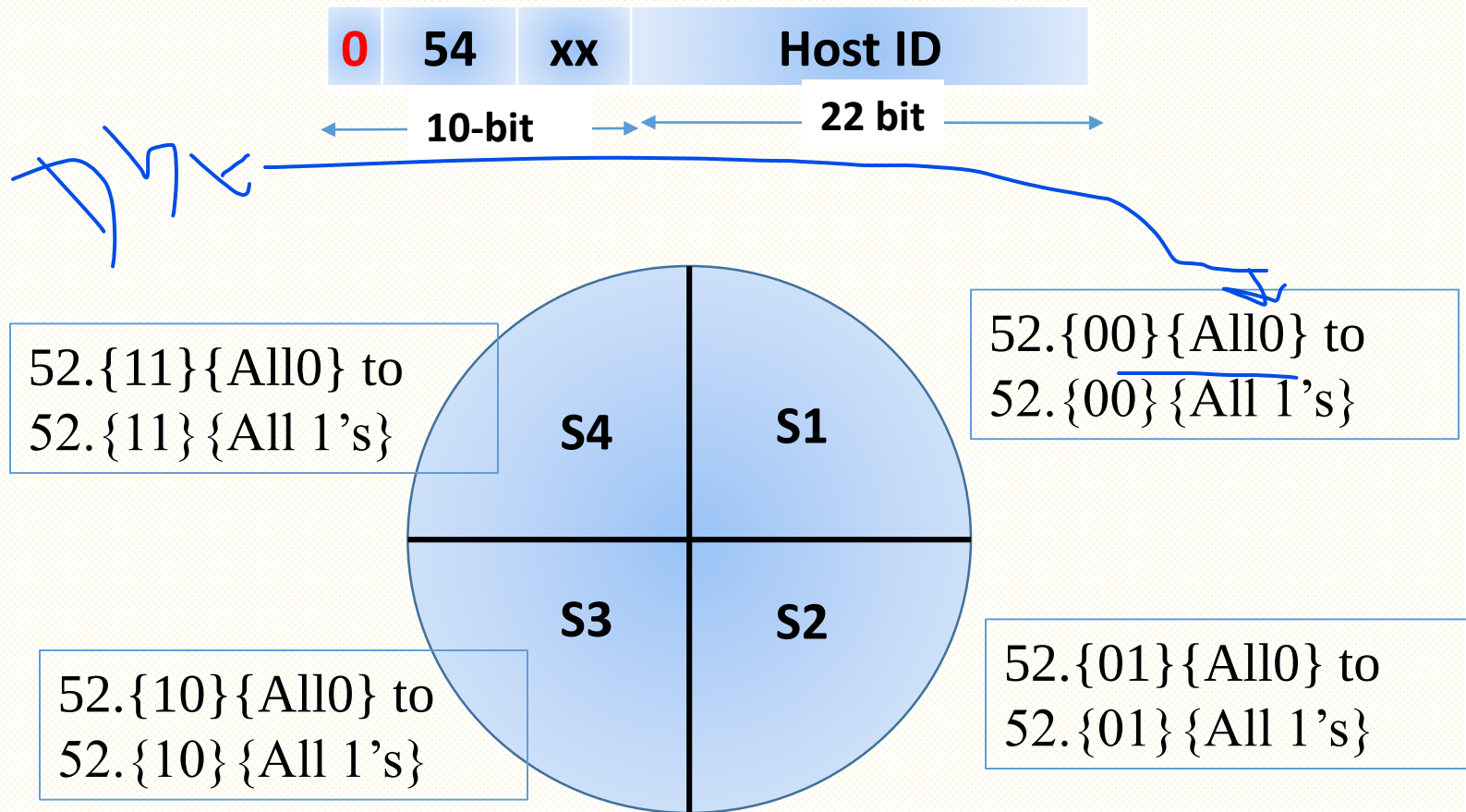


- ❑ 52.xx{0 to 64}.{0 to 255}.{0 to 255}



- ❑ xx → 00, 01, 10, 11 (sub networks with $2^{22} - 2$ host each)

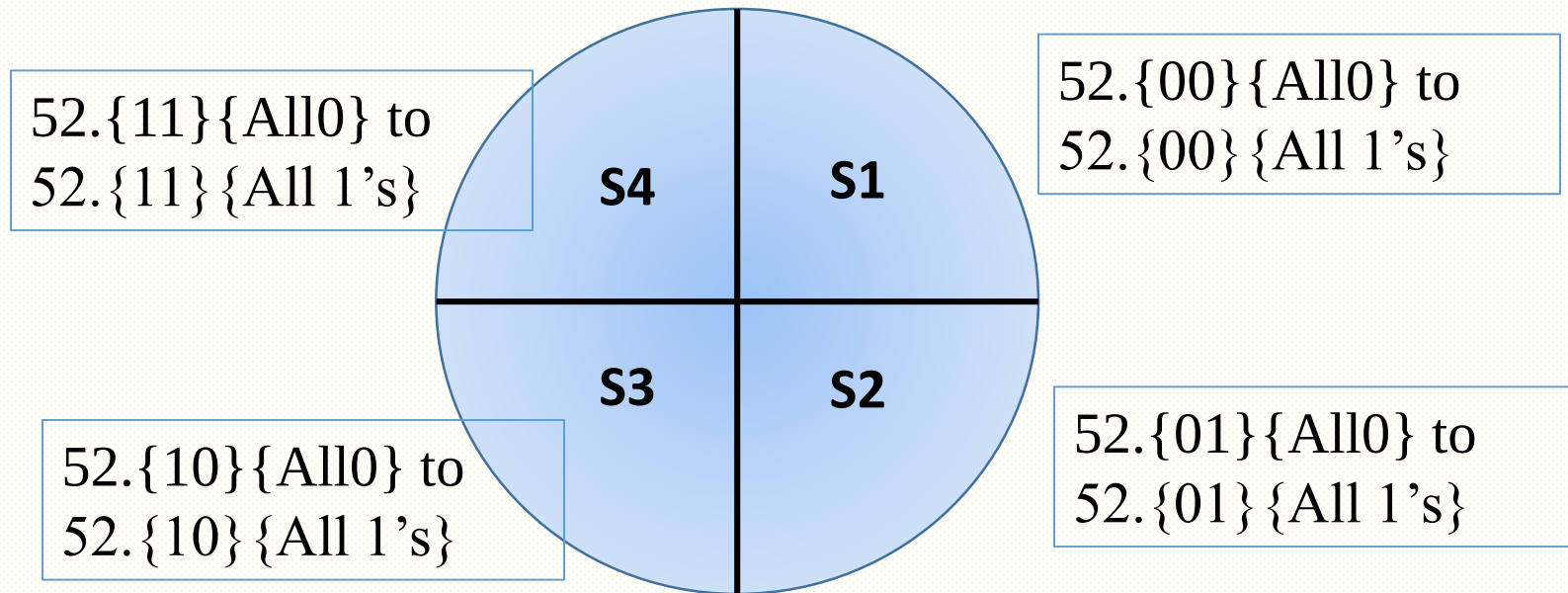
Sub-netting



❑ What will be broadcast and network address for each subnet?

Sub-netting

- ☐ How does the router know which subnet to send the packet?
- ☐ It will be handled by subnet mask.



- ☐ Subnet mask for this network will be?
- ☐ 255.192.0.0 .11

Sub-netting

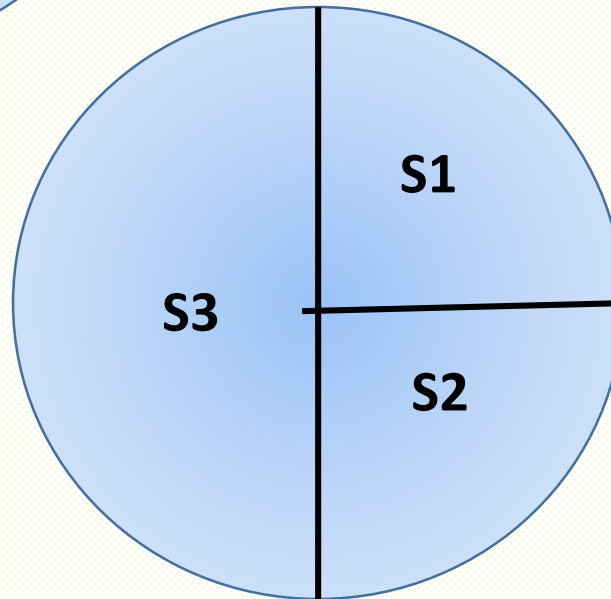
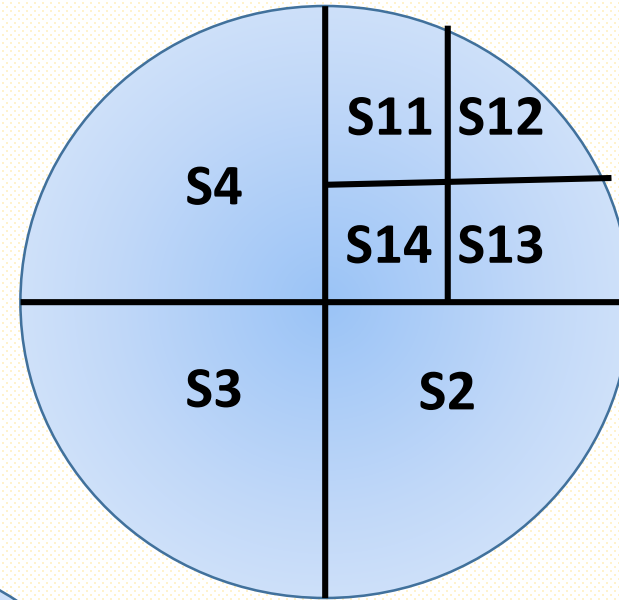
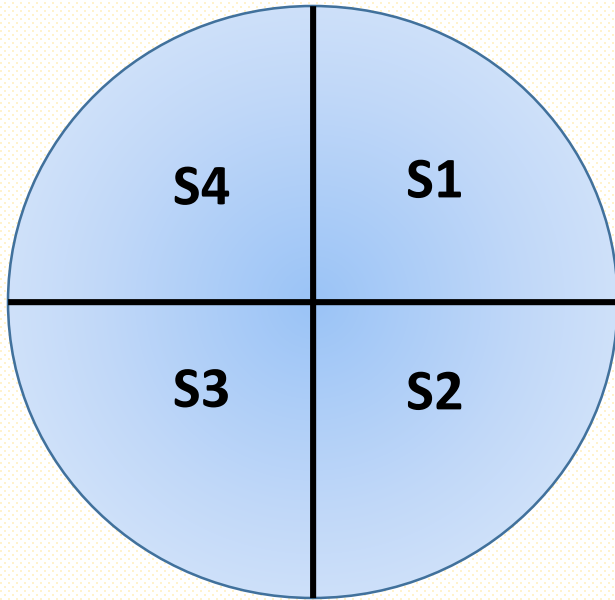


- ❑ How our routing table will look a like



Subnet Mask: 255.192.0.0		
52.0 – 52.63	S1	Eth0
52.64 – 52.127	S2	Eth1
52.128 – 52.191	S3	Eth2
52.192 – 52. 255	S4	Eth3
0.0.0.0	Default	

Hierarchical Sub-netting

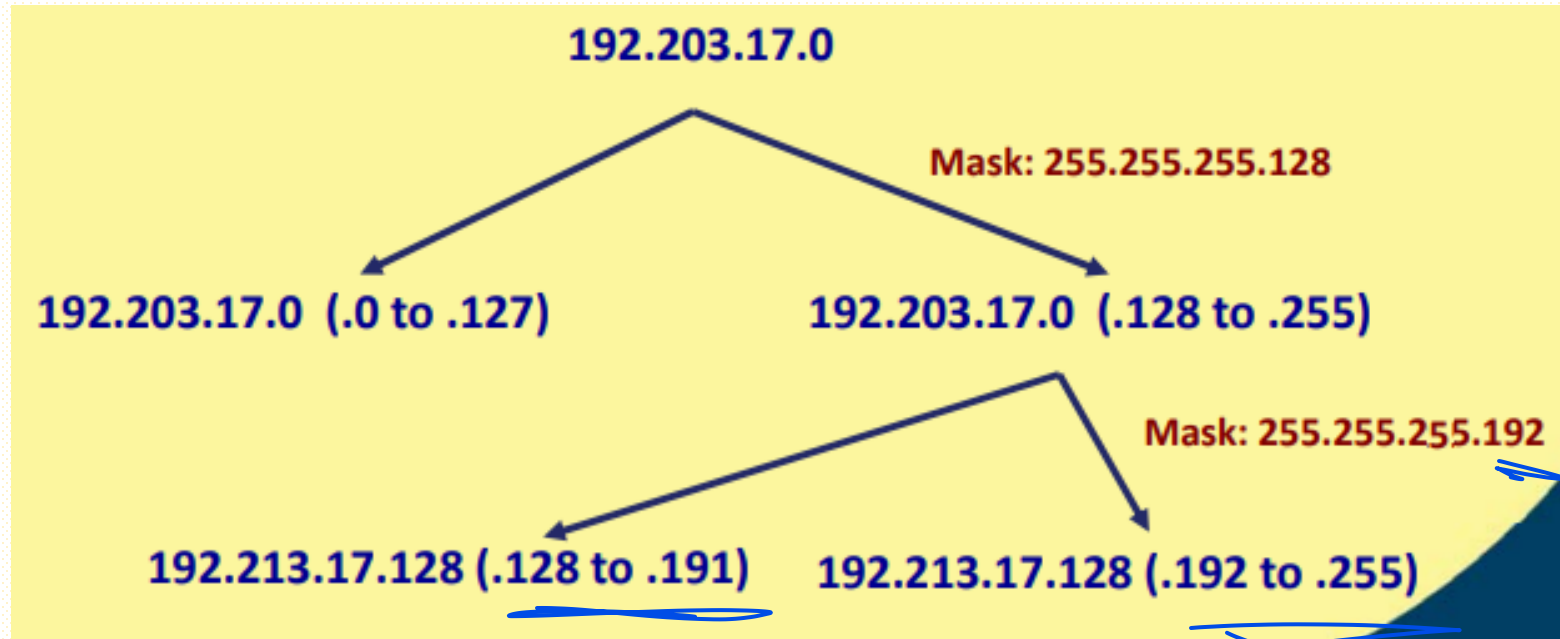


Classless Inter-Domain Routing (CIDR)



- ❑ No concept of class. It uses variable length subnet mask (VLSM)
- ❑ Assign any block of IP address to any ISP/company/Institute
- ❑ Allocate address based on actual need:
 - You need 50 host $\rightarrow$ x.x.x.x/26 (26 bit network id, 6-bit host)
 - You need 200 $\rightarrow$ x.x.x.x/24 (24 bit network id, 8-bit host)
- ❑ Company give requirement $\rightarrow$ IANA will assign IP range and tell the network and host bit.
- ❑ 10.10.1.32 to 10.10.1.63
- ❑ It is represented as **10.10.1.32/27**

Example: Subletting 192.203.17.0



Some Questions



❑ What is the subnet address if the destination IP address is 144.16.34.124 and the subnet mask is 255.255.240.0?

- 144.16.34.124 → 10010000.00010000.00100010.01111100
- 255.255.255.240.0 → 11111111.11111111.11110000.00000000
- Bitwise AND → 10010000.00010000.00100000.00000000
- 144.16.32.0 (Subnet address)

- ❑ An entry in the routing table has 135.46.56.0 as the destination and /22 as the subnet mask. What will be the network address?
- 135.46.56.0
 - /22 means 22 bits as network ID, 10 bit host.
 - Subnet mask: 11111111.11111111.11111100.00000000 (255.255.252.0)
 - IP : 10000111.00101110.00111000.00000000
 - Net. Addr : 10000111.00101110.00111000.00000000 (135.46.56.0)

- ❑ An entry in the routing table has 155.86.58.123 as the destination and /22 as the subnet mask. What will be the network address?
- /22 means 22 bits as network ID, 10 bit host.
 - Subnet mask: 11111111.11111111.11111100.00000000 (255.255.252.0)
 - IP : 10011011.01010110.00111010.01111011
 - Net. Addr : 10011011.01010110.00111000.00000000 (155.86.58.0)

Routing Table Questions

- ❑ Consider the following routing table in a router. On which interface will an IP packet with destination address 144.25.64.120 be forwarded?

Destination	Subnet Mask	Interface
144.25.0.0	255.255.0.0	Eth0
144.25.64.0	255.255.224.0	Eth1
144.25.68.0	255.255.255.0	Eth2
144.25.68.64	255.255.255.224	Eth3
default	0.0.0.0	Eth2

1 entry
match

- ❑ 144.25.64.120 $\rightarrow$ x.x.01000000.01111000
- ❑ Check against first entry 144.25.0.0/16 (match 144.25.0.0 to 144.25.255.255)

Routing Table



- ❑ Consider the following routing table in a router. On which interface will an IP packet with destination address 144.25.64.120 be forwarded?

Destination	Subnet Mask	Interface
144.25.0.0	255.255.0.0	Eth0
144.25.64.0	255.255.224.0	Eth1
144.25.68.0	255.255.255.0	Eth2
144.25.68.64	255.255.255.224	Eth3
default	0.0.0.0	Eth2

- ❑ 144.25.64.120 $\rightarrow$ x.x.01000000.01111000
- ❑ Check against second entry 144.25.64.0/19 (match 144.25.64.0 to 144.25.95.255)

Routing Table

- ❑ Consider the following routing table in a router. On which interface will an IP packet with destination address 144.25.64.120 be forwarded?

Destination	Subnet Mask	Interface
144.25.0.0	255.255.0.0	Eth0
144.25.64.0	255.255.224.0	Eth1
144.25.68.0	255.255.255.0	Eth2
144.25.68.64	255.255.255.224	Eth3
default	0.0.0.0	Eth2

- ❑ 144.25.64.120 $\rightarrow$ x.x.01000000.01111000
- ❑ Check against third entry 144.25.68.0/24 (range 144.25.68. 0 to 144.25.68.255) no match

Routing Table

- ❑ Consider the following routing table in a router. On which interface will an IP packet with destination address 144.25.64.120 be forwarded?

Destination	Subnet Mask	Interface
144.25.0.0	255.255.0.0	Eth0
144.25.64.0	255.255.224.0	Eth1
144.25.68.0	255.255.255.0	Eth2
144.25.68.64	255.255.255.224	Eth3
default	0.0.0.0	Eth2

- ❑ 144.25.64.120 $\rightarrow$ x.x.01000000.01111000
- ❑ Longest prefix match /19, so eth1.

Routing Table



- ❑ Consider the following routing table in a router. On which interface will an IP packet with destination address 161.44.64.120 be forwarded?

Destination	Subnet Mask	Interface
161.44.0.0	255.255.0.0	a
161.44.64.0	255.255.224.0	b
161.44.68.0	255.255.255.0	c
161.44.68.64	255.255.255.224	d
default	0.0.0.0	e